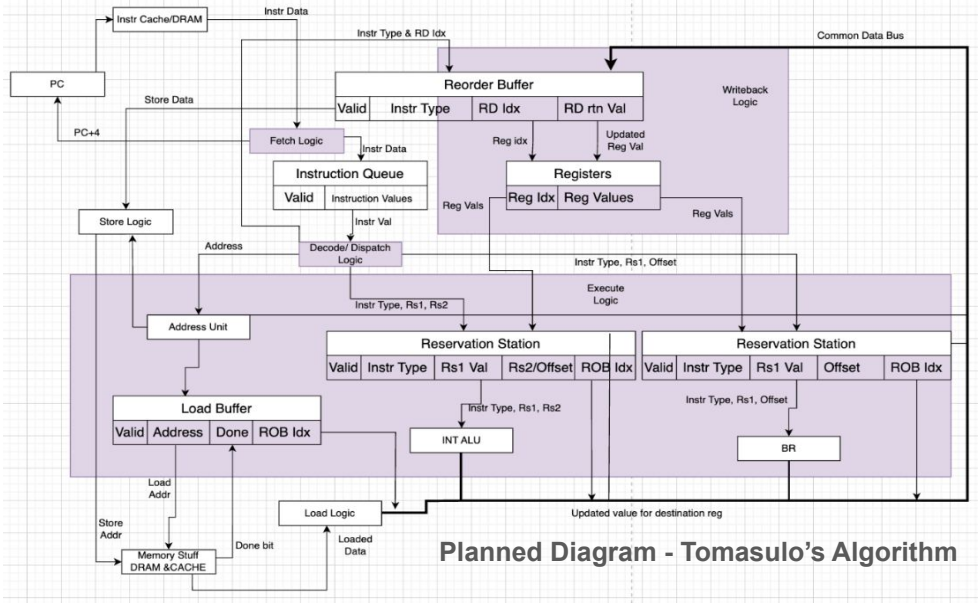
I. CP3 Case Design Logistics

Major problems and goals

- Low IPC and high delay
  - IPC to improve as we add new features e.g. branch predictors, prefetcher
  - Reduce slack to allow higher clock frequency
- Reservation stations depth is small
  - Requires prefetching to make use of RS & ROB space
- Lots of combinational logic in always_ff statements
  - Resolving these will improve timing & data flow
- Minimal increase in area desired



Benchmark	IPC	Delay (ps)
coremark	0.210199	13,870,820,000
aes_sha	0.192321	40,345,330,000
compression	0.216332	22,273,410,000
fft	0.192158	60,419,880,000
mergesort	0.228405	18,388,980,000

```
Clock frequency = 100MHz
data required time      9.958930
data arrival time             -8.720723
-------------------------------------
slack (MET)                    1.238207
```

```
Combinational area:     71657.474373
Buf/Inv area:           5322.659996
Noncombinational area:  60164.144060
Macro/Black Box area:   79531.081055
Net Interconnect area:  undefined
-------------------------------------
Total cell area:        211352.699487
```

II. Advanced Features & Optimizations

Next-Line Prefetcher - A module that interfaces with the cache.

Post-Commit Store Buffer - A ring buffer that merges data for existing entries, and forwards to loads.

Age-Ordered Issue Scheduling - Uses queues where the head was the most likely oldest station, looping to the next ready station.

Compression extension - Has a secondary decode for compressed instructions, in parallel with normal decode.

Tournament predictor with GSelect and two-level - Uses a 4 bit saturating counter to decide between the two of them, which is incremented/decremented when their opinions differ.

Split LSQ - uses separate queues for load and store rs stations. Check for the next oldest load instruction younger than store through checking order.

Benchmarkz	IPC	DP^2
coremark	0.4131	37283.25
aes_sha	0.4077	1953559.83
compression	0.5678	49858.68
fft	0.4251	8243623.53
mergesort	0.4221	256849.65

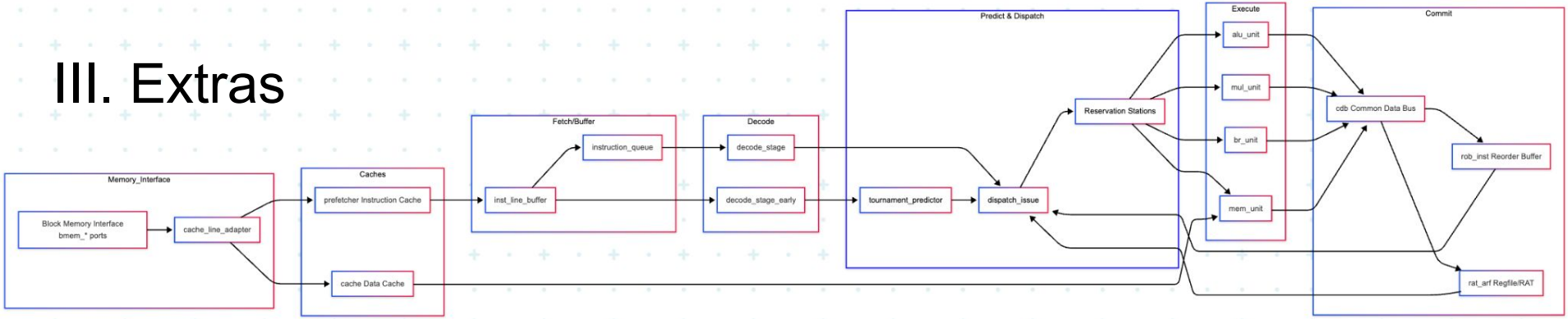
Branch Statistics for coremark_im

	Number of branches	Number of flushes	IPC
Static not taken	111,945	34,821	0.335
Tournament	75,303	16,899	0.413

Prefetcher Statistics for coremark_im

	Cycles in compare state	Hits in compare state	Percent Cache Hits
With Prefetcher	203,157	199,869	98.3%
Without Prefetcher	102,586	99,493	96.9 %

III. Extras



To decrease critical path age-ordered scheduling was changed from comparing order of all rs stations, to finding the most likely eldest rs station by forming a queue like structure.

Area limitations required us to reduce the size of all reservation stations from 8 to 4 except for the ALU. This actually allowed an increase in IPC, maybe due to improving the chances of finding the oldest station to execute since it's not based strictly on order.

Certain combinations of the advance features did not work well together. For instance, compression extension was implemented in a separate branch to save area.

Improvements from earlier implementation:
Reservation station - increase the depth and allow adding dispatched instruction even when the functional unit is executing an instruction. Out-of-order issue enabled.
Stall - removed stalling every other cycle intended to block preemption for executing functional units, including memory instructions.

We used VSCode's Liveshare to collectively resolve Lint/Synthesis warnings and separate branches for developing new features.